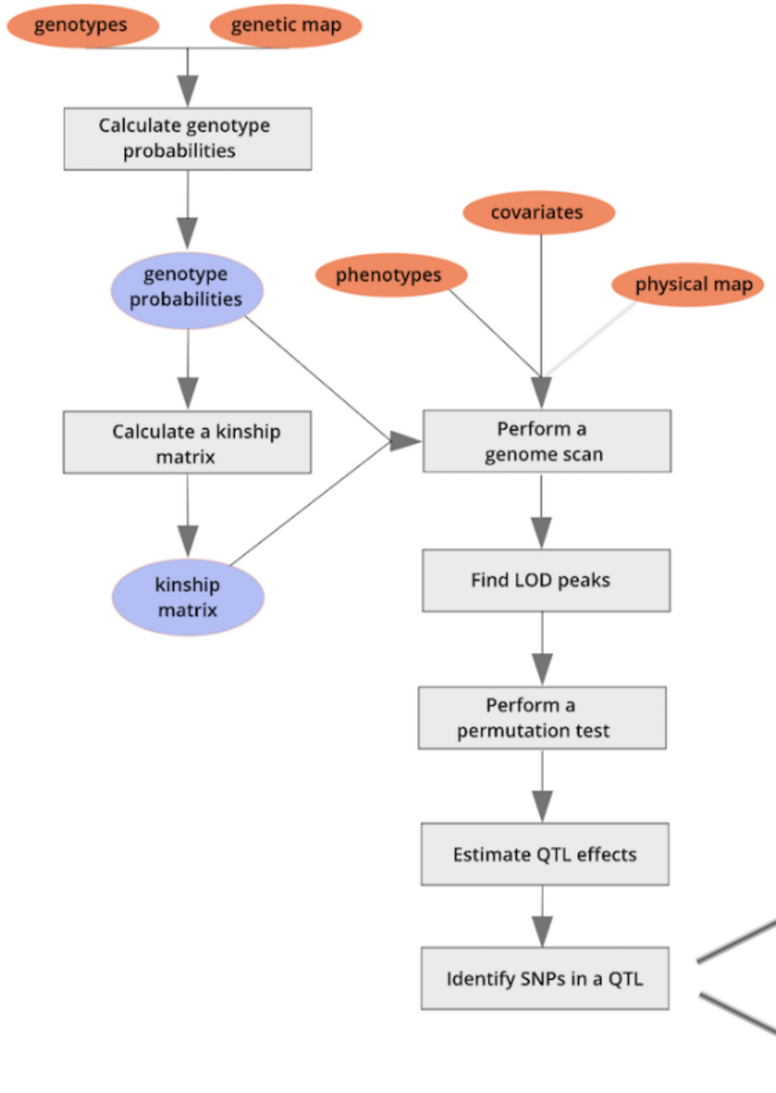
Pheno2GeneRank

Introduction: Biological Background of GWAS or QTL



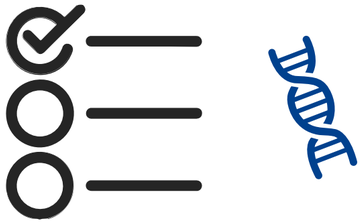
"Goal: To prioritize causal genes within QTL that is driving certain trait / phenotype"

Trait / Phenotype





Candidate Gene List





Gene Exploration & Selection

Previous tools (QTG_Finder, QTG-LGBM)

Feature Engineering
→ Machine Learning

Use only predefined features

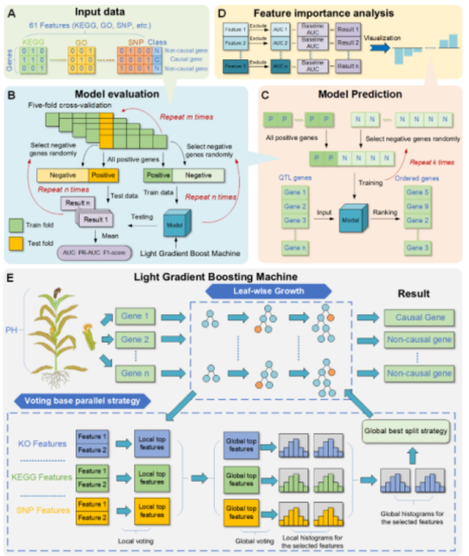
Do not use
trait / phenotype context

QTG_Finder (version 2.0)

QTG-Finder is a machine-learning pipeline to prioritize causal genes for QTLs identified by linkage mapping. We trained QTG-Finder models for Arabidopsis, rice, sorghum, Setaria viridis based on known causal genes and orthologs of known causal genes, respectively. By utilizing additional information like polymorphisms, function annotation, co-function network, paralog copy number, the models can prioritize causal genes for QTLs identified by QTL mapping.

Authors: Fan Lin, February 2020
Environment: Python=3.7.12 pandas=0.25.3 numpy=1.19.5 scikit-learn=0.21.2

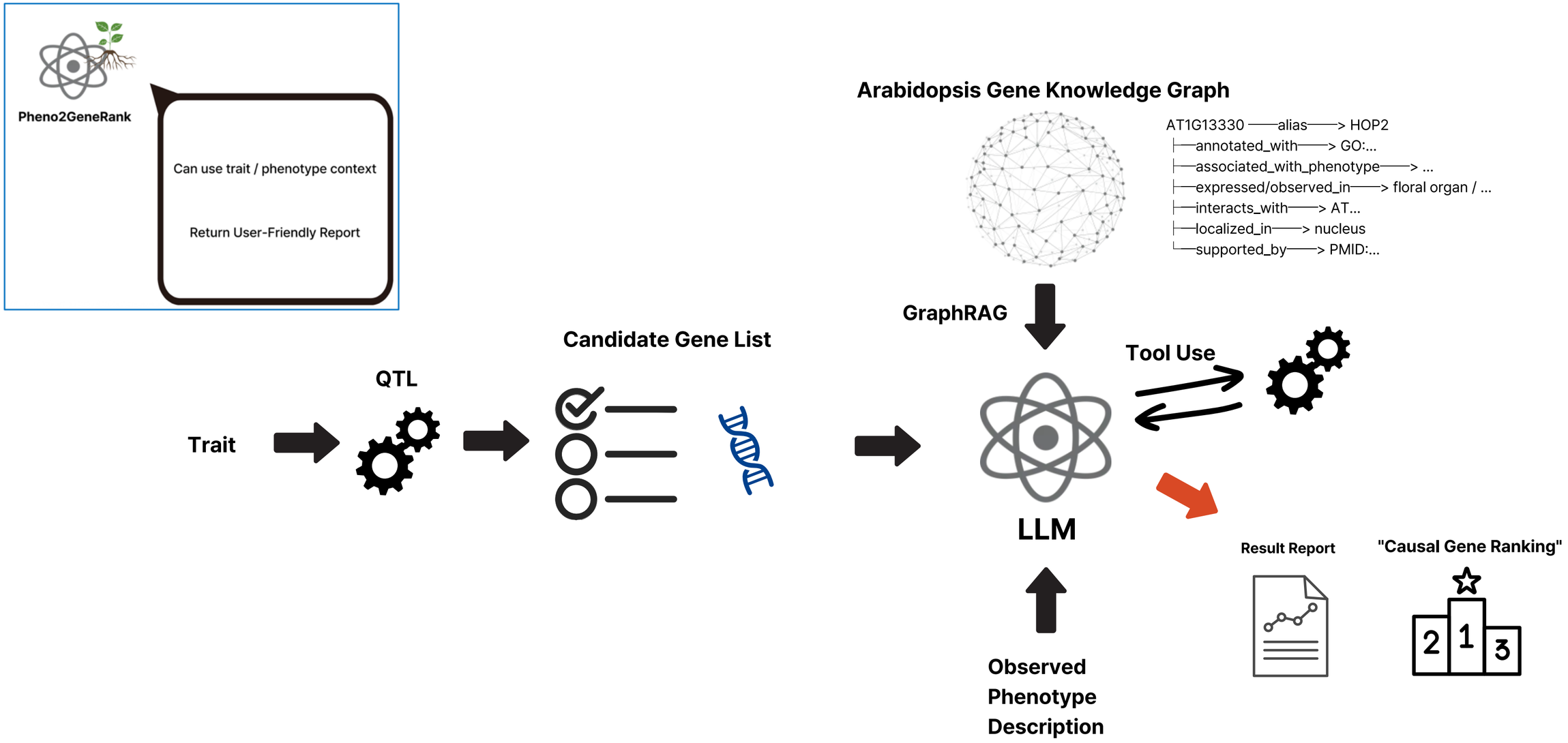
QTG-LGBM: a method of prioritizing causal genes in quantitative trait loci in maize



QTG-Finder : employs random forest (RF) to predict causal genes within flowering time (FT)-related QTL in Arabidopsis thaliana (Arabidopsis), *Oryza sativa* (rice), Setaria *viridis* (Setaria), and *Sorghum bicolor* (sorghum)

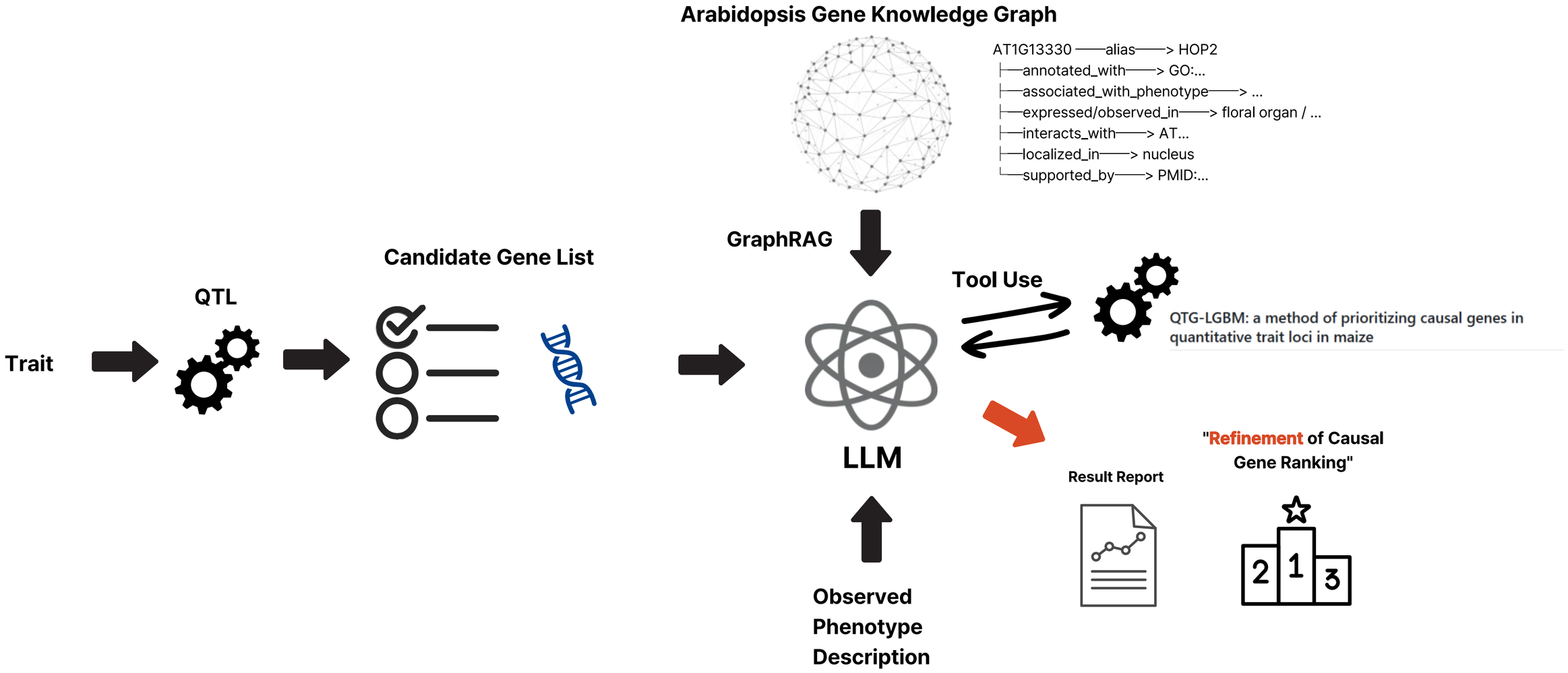
The input for ***QTG-LGBM*** involves 61 features about 25,629 genes (Fig. 1A). These gene features comprise 17 Kyoto Encyclopedia of Genes and Genomes (KEGG) features, 17 Gene Ontology (GO) features, and 27 Single Nucleotide Polymorphism (SNP) features.

Idea of Pheno2GeneRank: Causal Gene Ranking LLM Agent through Observed Phenotype Descriptions

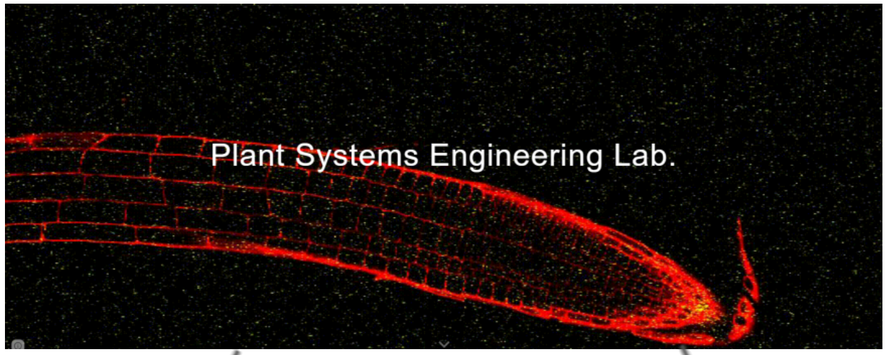


Idea 2 of Pheno2GeneRank: Use QTG-LGBM as a tool for Pheno2GeneRank

→ We can use both the predefined features (that are used for QTG-LGBM) and phenotype context



"Cowork"



Supervision
for Pheno2GeneRank

Plant System Research with Pheno2GeneRank



Systems Biology and Bioinformatics Laboratory

Pheno2GeneRank Development

1. Algorithm Design
2. Ablation Study

Evaluation

1. Metric Design
2. Test Dataset Construction
3. Validation

Step-by-step workflow

- Step1. Collect input data for testing
  - Which trait will be collected?
  - What kind of gold standard (known causal genes from QTL experiments) are available?

- Step2. Build LLM Agent
  - What kind of LLM agents will be constructed (idea 1 vs. idea 2)?
  - What will be the output (what will provide in the report)?
  - What will be the evaluation metric?

- Step3. Agent interpretation / Feature importance analysis / Experimental validation
  - Can we provide the reason why certain gene is causal?
  - How can we experimentally validate causal gene?